

CASE REPORT

Pathology/Biology

Unexpected death in a young man associated with a unilateral swollen leg: Pathological and toxicological findings in a fatal snakebite from *Deinagkistrodon acutus* (Chinese moccasin)

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Abstract

Deinagkistrodon acutus (*D. acutus*), also known as the Chinese moccasin, is a viper species found throughout the southeastern parts of China, northern Vietnam and Laos. *D. acutus* envenomation can result in coagulopathy and lead to death if not treated correctly. A 20-year-old man was discovered with a severely swollen left thigh with overlying dark purple, discolored skin. He was immediately transported to hospital. Laboratory examinations revealed dysfunctional coagulation and fluid-electrolyte imbalances. He died 2 h later despite resuscitation efforts. Surveillance footage revealed that he had walked through a grass field while returning home that night. Autopsy and pathological examination findings revealed a large area of muscle necrosis of the left thigh, renal tubular necrosis, and hepatocyte necrosis. Potential fang marks were found on the decedent's jeans. Due to our suspicions, we performed specific enzyme-linked immunosorbent assays (ELISA) and detected *D. acutus* venom in the kidneys, left thigh muscle, liver, lung, spleen, and heart tissues of the decedent. In conclusion, the clinical manifestations, autopsy, histopathological examination, ELISA, and investigation results confirmed *D. acutus* envenomation.

KEYWORDS

forensic pathology, forensic toxicology, snakebite, *Deinagkistrodon acutus*, ELISA, autopsy, Chinese moccasin, venom, China

1 | INTRODUCTION

Snakebite envenomation is an important global health consideration. Snakebites lead to approximately 2.4 million envenomations and 94,000–125,000 deaths annually. In addition, snakebites account for an additional 400,000 amputations, as well as other severe sequelae [1]. The highest incidence occurs in South Asia, south-east Asia, and sub-Saharan Africa [2]. Bolon et al. reviewed 33,827 snakebite cases across 35 countries. Approximately 70% of these victims, or witnesses to the event, recognized the specific snake species,

while 38% of them brought a captured or dead snake to the hospital. However, only 53% of these snakebite cases were identified at species/genus level [3].

Deinagkistrodon acutus (*D. acutus*) belongs to the *Viperidae* family, *Crotalinae* subfamily and *Deinagkistrodon* genus (containing only *D. acutus*), with widespread distribution throughout the south-eastern parts of China, northern Vietnam, and Laos [4]. The mortality rate of *D. acutus* envenomation remains high in China, at approximately 24% [5].

Snake venom is a complex mixture. The toxins of significance in cases of human envenomation include neurotoxins, cardiotoxins,

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hemotoxins, and cytotoxins. The constitution of snake venoms differs between species [6].

In forensic practice, death due to envenomation is usually diagnosed based on a clear history, capture of the offending snake, presence of fang marks, and characteristic clinical manifestations with gross pathological changes. Nevertheless, in cases with an absence of a clear snakebite history, toxicology is critical in making an accurate diagnosis. ELISA is a widely used detection method for snake toxins because of its high sensitivity, specificity, and its convenience in the quantitation of biotoxins with high molecular weights [7].

Here, we report a fatal *D. acutus* envenomation case confirmed by ELISA in which fang marks were unidentified on the body.

2 | CASE REPORT

2.1 | Case history

A 20-year-old man, still in bed, was found by his roommate at approximately 8:00 a.m. with a swollen left thigh and dark purple, discolored skin. He was already in a coma and had difficulty breathing, and was rapidly transported to the hospital. His pupils were dilated. Initial laboratory examinations revealed pH 6.80, Na⁺123 mmol/L (135–145), K⁺ >9 mmol/L (3.5–5.5), platelets (PLT) 123×10^9 /L (125–350), C-reactive protein (CRP) 168 mg/L (0–10), D-dimer 4.79 mg/L (0–0.55), and fibrinogen 0.18 g/L (2–4). His blood did not clot during coagulation tests. Blisters were observed on the skin of his left thigh, with the largest being 3.0 × 2.0 cm in size. Despite resuscitation, he was declared dead at 11:00 a.m. that morning. According to additional police investigations, he had had dinner with friends and gotten drunk the evening before his death. Surveillance footage revealed that he had walked through a grass field while returning home that night.

2.2 | Autopsy findings

A forensic autopsy was performed within 24 h. Conjunctival petechiae were observed bilaterally. Bloody secretions were observed

Highlights

- A fatal snake envenomation case caused by *D. acutus*, with the absence of fang marks on the body, was reported with autopsy and pathological findings.
- Examination of the decedent's clothing provided clues to a snakebite history.
- ELISA test confirmed the existence and quantified the amount of *D. acutus* venom in the decedent's kidneys, left thigh muscle, liver, lung, spleen, and heart samples.
- A review of *Deinagkistrodon acutus* envenomation cases and a review of common venomous snakes in the region and the consequences of their bites are presented in the Tables.

in the bilateral nasal cavities and the oral cavity. There was 300 and 50 mL of hemorrhagic fluid in the left and right chest cavities, respectively. A small amount of hemorrhagic fluid was found in the abdominal and pelvic cavities.

The skin of the left portion of the lower body, from 5.0 cm below the umbilicus, down to the left ankle, including the penis and scrotum, was significantly swollen with a dark purple discoloration. There was a clear demarcation of the left, lower affected area (Figure 1A). Subcutaneous crepitus was felt when palpating the affected area. Multiple blisters varying in size were noticeable over this area (Figure 1B). The epidermis of the blisters was easy to peel off, exposing the underlying dermis. There were also small areas of bruising and scratches scattered over the surface of the skin on the left thigh.

A large amount of light red liquid drained when these blisters were incised. Extensive hemorrhage, necrosis, and edema were found in the subcutaneous soft tissue and muscles of the affected area. The medial portion of the left thigh muscles revealed significant necrosis, while the lesions on the lateral part of the left lower extremity were relatively mild in comparison (Figure 2A). No fang

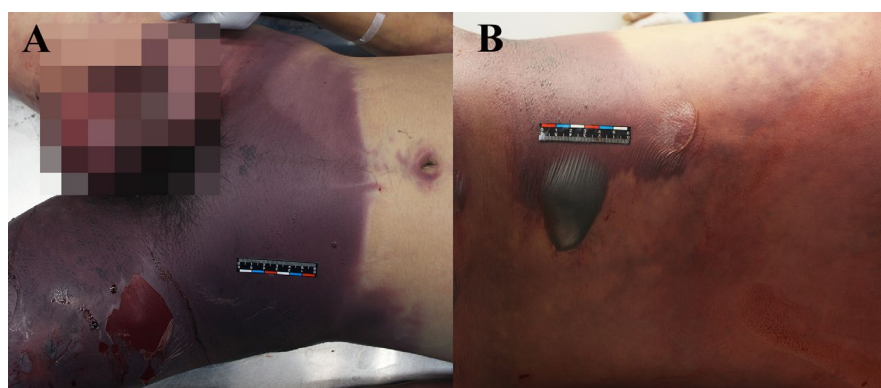


FIGURE 1 (A) The lesion area was swollen, clearly demarcated by dark purple discoloration (B) Blisters on his left waist

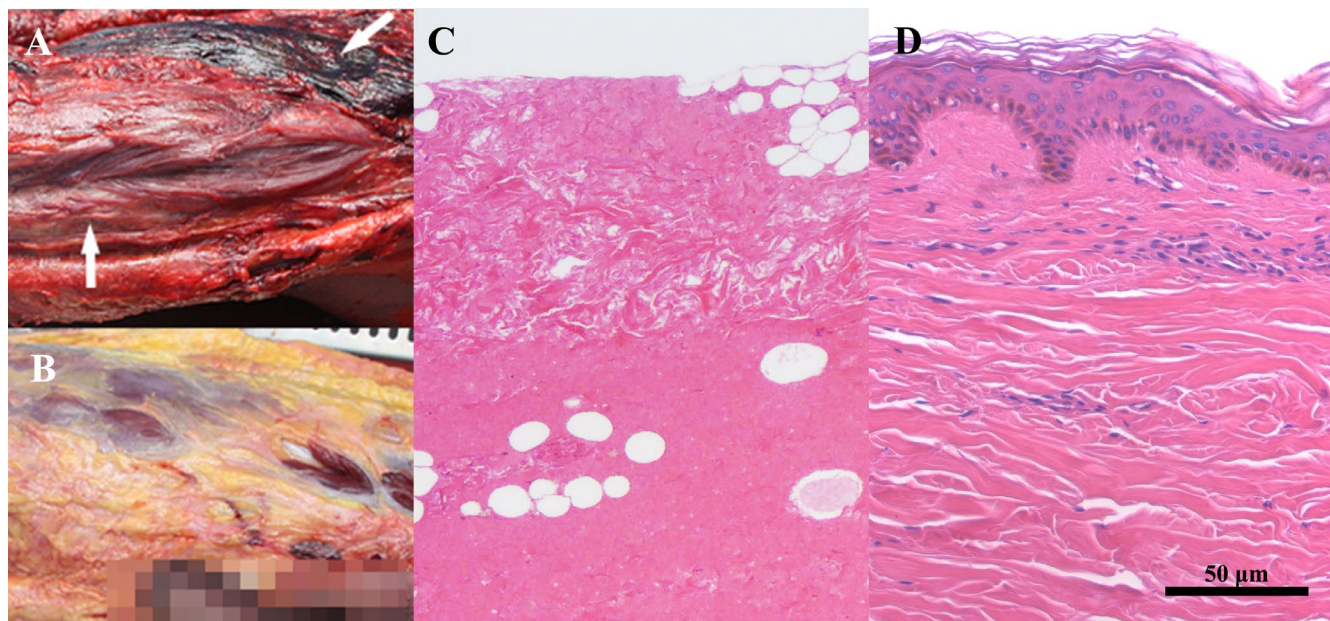


FIGURE 2 (A) The muscles of the medial (↙) portion of the left thigh showing severe necrosis, the muscle lesions of the lateral (↑) portion were relatively mild. (B) Normal appearance of the right thigh. (C) Hemorrhage, edema, and necrosis of the left medial thigh tissue. (D) The skin of the right thigh remained normal

marks or puncture sites were observed despite careful inspection of the area. In contrast, the skin, subcutaneous soft tissue and muscles of the right thigh were normal in appearance (Figure 2B).

2.3 | Pathologic findings (gross and microscopic pathology)

In H&E sections prepared from the skin of the left thigh, the epidermis could not be observed due to blistering. Several bubbles containing exudate were noted in the dermis and subcutaneous areas. Massive hemorrhaging was observed subcutaneously (Figure 2C). The pathological changes observed in the medial thigh tissues of the left leg were more significant than those of the lateral thigh tissues of the same leg, while the skin of the right thigh remained normal (Figure 2D).

Both kidneys together weighed 320 g. Both kidneys were swollen and congested. Microscopically, postmortem autolysis was significant, while tubular epithelial cells showed degeneration and necrosis. Hyaline and granular casts were found in the lumen of the renal tubules.

Multiple patches of scattered hemorrhaging were observed on the surface of the liver. A wide range of coagulative necrosis of the hepatocytes and infiltration of mononuclear cells were observed.

Other pathologic findings included pulmonary edema and hemorrhage, moderate cerebral edema, mild neuronal degeneration/necrosis, myocardial contraction band necrosis, splenic congestion and hemorrhage, adrenal congestion, and thyroid congestion. No thrombosis was found in the major vessels.

Organ and tissue samples obtained at autopsy were frozen at -80°C prior to sandwich-ELISA testing.

2.4 | Investigation

Further investigation was requested, and the police provided the jeans that the deceased was wearing on the evening before his death. After careful inspection of the jeans, 2 spots of blood approximately 3.0 cm apart were discovered on the left anterior thigh portion of the jeans (Figure 3). This location coincided with the anatomical location that exhibited the most severe pathological changes on the victim.

2.5 | Laboratory examination

2.5.1 | Sandwich-ELISA test

Sandwich-ELISA was performed to detect the venom of various snakes including *D. acutus*, *Agkistrodon halys*, *Trimeresurus stejnegeri* (*T. stejnegeri*), and *Naja atra* as described [8]. These species were chosen based on the pathological changes observed and the geographic epidemiological features of snakebites [9].

Materials: Horse anti-*D. acutus* polyclonal antibody (PcAb), horse anti-*Agkistrodon halys* PcAb, horse anti-*T. stejnegeri* PcAb, horse anti-*Naja atra* PcAb (Serum, China). Snake venom (all in powder) of *D. acutus*, *Agkistrodon halys*, *Trimeresurus stejnegeri*, and *Naja atra* (Kunming Institute of Zoology, Chinese Academic of Science) were included. Phosphate-buffered saline, pH 7.4 containing 0.05% Tween-20 (PBST), 0.1 M sodium carbonate-bicarbonate, pH 9.6 (CBS), bovine serum albumin (BSA), ddH₂O, tetramethylbenzidine (TMB), and 2.0 M H₂SO₄ were used.

Methods: Part of the PcAb was made into HRP-labeled snake venom antibody using the NaIO₄ method. Then, polystyrene



FIGURE 3 Two hemorrhagic spots 3.0 cm apart were found on the left, inner area of the jeans

micro-ELISA plates were coated with snake venom PcAb overnight at 37°C. After washing and 4–5 h of blocking (2% BSA in PBST), the plates were incubated at 37°C overnight. 100 microliters of diluted snake venom (or organ homogenate) in sample buffer (1% BSA in PBST) was delivered into each well and incubated for 60 min at room temperature. The plate was washed, and bound snake venom was detected with HRP-labeled snake venom antibodies. Enzymatic activity was measured with the addition of TMB. The color appeared in 10 min or less at 37°C in darkness. The reaction was terminated by the addition of sulfuric acid, and the absorbance at 450 nm was measured with a microplate reader (Thermo Scientific).

The left thigh muscle, kidney, heart, spleen, lung, and liver samples were homogenized for this test.

Results: *D. acutus* venom was detectable and quantified in all the organ samples tested: kidneys 13.9 µg/g, left thigh muscle 10.5 µg/g, liver 6.62 µg/g, lung 4.93 µg/g, spleen 2.50 µg/g, and heart 1.53 µg/g.

2.5.2 | Toxicology tests

Toxicologic analysis revealed a blood ethanol concentration of 0.0088 gm/dL and no other detected compounds.

2.5.3 | Microbial cultures

Results of microbial cultures from a blood sample obtained from the heart (aerobic and anaerobic culture): *Escherichia coli*, *Streptococcus salivarius*, *human Staphylococcus hominis subspecies*, and *Streptococcus oralis* (+).

2.6 | Diagnosis

The examinations and tests indicated that the victim died from *D. acutus* envenomation. The venom caused massive areas of hemorrhage and global necrosis in the left lower limb, waist, and back. This

resulted in electrolyte imbalances and eventually led to irreversible shock and multiple organ failure.

3 | DISCUSSION

D. acutus venom is primarily composed of hemotoxins. According to proteomic research reported in 2018 and 2019, *D. acutus* venom is composed of 4 families of proteins: snake venom metalloproteinase (SVMPs), snake venom C-type lectin (CLECs), phospholipase A₂s (PLA₂s), and snake venom serine protease (SVSPs) [10,11].

SVMPs are abundant in *D. acutus* venom and are capable of degrading most of the constituents in the extracellular matrix and causing coagulopathy, which explains local and systemic hemorrhagic symptoms such as ecchymosis, blisters, and hemorrhagic bullae. Most of the CLECs in *D. acutus* venom act as antagonists of platelet aggregation causing thrombocytopenia. We reviewed the reported case of *D. acutus* envenomation in Table 1 [4,12–24]. 82.29% (302/367) of these cases showed abnormalities in platelet counts. PLA₂s in *D. acutus* venom show anticoagulant activity, while SVSPs are plasminogen activators catalyzing plasmin activity via plasminogen proteolysis [10]. Accordingly, *D. acutus* envenomation cases revealed abnormalities in coagulation tests, such as prothrombin time (PT), activated partial thromboplastin time (APTT), FIB (fibrinogen), and D-dimers, as shown in Table 1. These active components are closely associated with clinical manifestations including hemorrhage, hypofibrinogenemia, coagulopathy, and thrombocytopenia [13], which were all observed in the decedent described in this case. These symptoms are reversible if specific antivenom and other medical therapies are administered in time [13]. Unfortunately, the decedent was not aware of the bite because of alcohol intoxication.

The identification of snakebite cases depends on the history, characteristic fang marks, and the snake being captured. Fang marks are located primarily on the extremities of the victims (Table 2). Unusual fang mark locations could indicate unnatural envenomation seen in intentional envenomation or murder [25,26]. Nevertheless, in a review of 129 *D. acutus* envenomation cases, clear fang marks were only observed in 89 cases [17]. Due to the severe damage of the left leg of the decedent, the original fang marks were unidentified. However, after examining the jeans of the victim, we were able to clearly identify 2 spots of blood on the left part of the jeans, overlaying the area of skin that was most severely damaged. This evidence was pivotal in this case, as it supports our diagnosis of a snakebite, despite not being able to observe the characteristic fang marks directly on the victim's body. The expected distance between the fang punctures of *D. acutus* is 1.5–3.5 cm according to literature, relatively large among snakes [27]. The distance in this case was 3.0 cm, in accordance with previous research. In this case, we measured the distance between the centers of the 2 spots. The well-defined upper limit of the discoloration, appearing as a band across the lower abdomen, was also noticeable. The phenomenon could be explained by 2 reasons: Firstly, the decedent had developed severe

TABLE 1 Review of *D. acutus* envenomation cases.

Location of bite [4,12-24]	Occurrence rate	Local manifestations [4,12-14,16-24]	Other manifestations [4,12-24]	Coagulation function examination (abnormal%) ^a
Upper Limbs 38.36% (239/623)		Swelling 100% (347/347)	Neurological Symptoms Internal or Mucosal Hemorrhage	PT 27.03% (40/148) [12-13,17,19,24]
Left	26.36% (63/239)	Blisters or Hemorrhagic Bullae	Acute Renal Failure Cyanosis	APTT 25.85% (38/147) [4,13,17,19,24]
Right	57.32% (137/239)	13.26% (46/347)	Infection	PLT 82.29% (302/367) [4,13,15,17-20]
Not Mentioned	16.32% (39/239)	Ecchymosis	DIC	FIB 27.40% (40/146) [4,12-13,17,19,24]
Lower Limbs 60.83% (379/623)		64.55% (224/347)	Shock	D-dimers 92.86% (13/14) [4,12,19]
Left	29.55% (112/379)	Bleeding	Compartment Syndrome	FDPs 100% (1/1) [13]
Right	25.86% (98/379)	49.57% (172/347)	Cerebral Infarction	
Not Mentioned	44.59% (169/379)	Necrosis	Acute Demyelinating Encephalo-myelitis	
Others	0.80% (5/623)	10.09% (35/347)		

Abbreviations: APTT, activated partial thromboplastin time (s); D, dimers ($\mu\text{g/L}$); FDPs, fibrinogen degradation products (mg/L); FIB, fibrinogen (g/L); PLT, platelet count ($\times 10^9/\text{L}$); PT, prothrombin time (s).

^aOnly results prior to antivenom treatments were included.

TABLE 2 Information of the 4 suspected snakes [5,30]

Specie of Snake	Reported Distribution of Snake	Pathologic Change	Mortality
<i>Deinagkistrodon acutus</i>	Distributed in the east of longitude 104° and between latitude 25° and 31°: Chinese provinces of Anhui, Zhejiang, Jiangxi, Fujian, Taiwan, Hubei, Hunan, Guangdong, Guangxi, Sichuan, and Guizhou; the north Vietnam	Swelling, ecchymosis, bleeding, tissue necrosis; facial, and visceral bleeding	24%
<i>Agkistrodon halys</i>	<i>Agkistrodon blomhoffii brevicaudus</i> : Chinese provinces of Hebei, southern Liaoning, Jiangsu, Anhui, Zhejiang, Jiangxi, Fujian, Taiwan, Hubei, Shaanxi (south of Qinling Mountains), Sichuan, and Guizhou; Korean peninsula <i>Agkistrodon blomhoffii ussuriensis</i> : Chinese provinces of Liaoning, Jilin, Heilongjiang, Inner Mongolia, Shandong, and Hebei; Korea and the far east area of Russia	Swelling, blisters, hemorrhagic bullae, ecchymosis; pupil diplopia, eyelid droop, dysphagia, dyspnea, hemoglobinuria	7%
<i>Trimeresurus stejnegeri</i>	Chinese provinces of Jiangsu, Anhui, Zhejiang, Jiangxi, Fujian, Taiwan, Henan, Hubei, Hunan, Guangdong, Hainan, Guangxi, Gansu, Sichuan, Guizhou, Yunnan, Jilin; Vietnam, Myanmar, Thailand, Bhutan, Sikkim, and India	Swelling, blisters, hemorrhagic bullae; dizziness or nauseous but the symptoms are relatively light compared with other poisonous snakes.	1%
<i>Naja atra</i>	Chinese provinces of southern Anhui, Zhejiang, Jiangxi, Fujian, Taiwan, Hubei, Hunan, Guangdong, Hainan, Guangxi, Sichuan, Guizhou, Yunnan; Thailand and Vietnam	blisters, hemorrhagic bullae, tissue necrosis, corestenoma	8%

swelling and the jeans were not elastic, so the clearly defined margin might well be the consequence of his restrictive clothing. Secondly, this portion of the left thigh was where the envenomation occurred and the superficial vein collecting blood in this portion is the great saphenous vein. The discoloration therefore correlates with the associated venous drainage pathways.

Capturing the offending snake is important for species identification. If it is unavailable, taking the geographic location into consideration and knowing the local incidence of snakebites for different species is crucial. Clinical manifestations are difficult to rely on and are more indicative of the type of toxins involved. In clinical or forensic practice, 3 types of detection methods are utilized to confirm the snake species responsible: genetic methods (PCR),

chromatography methods, or immunoassays [28]. Immunoassays are the most commonly used methods. ELISA has been modified and is the most widely used method for the detection of snake venom since 1977 [28,29]. ELISA plates coated with antibodies are usually pre-prepared and can be stored for more than 3 months [8]. This allows fast cross-activity detection of various types of snake venom and the quantitation of venom concentration in different samples within 2 hours.

Based on the distribution of venomous snakes [5,30] and the epidemiological features of snakebites in Hubei province, of all the snakebite cases identified (158), the top 4 species that could be responsible for the observed symptoms were *Agkistrodon halys* (70/158), *Naja atra* (62/158), *D. acutus* (6/158), and *Trimeresurus*

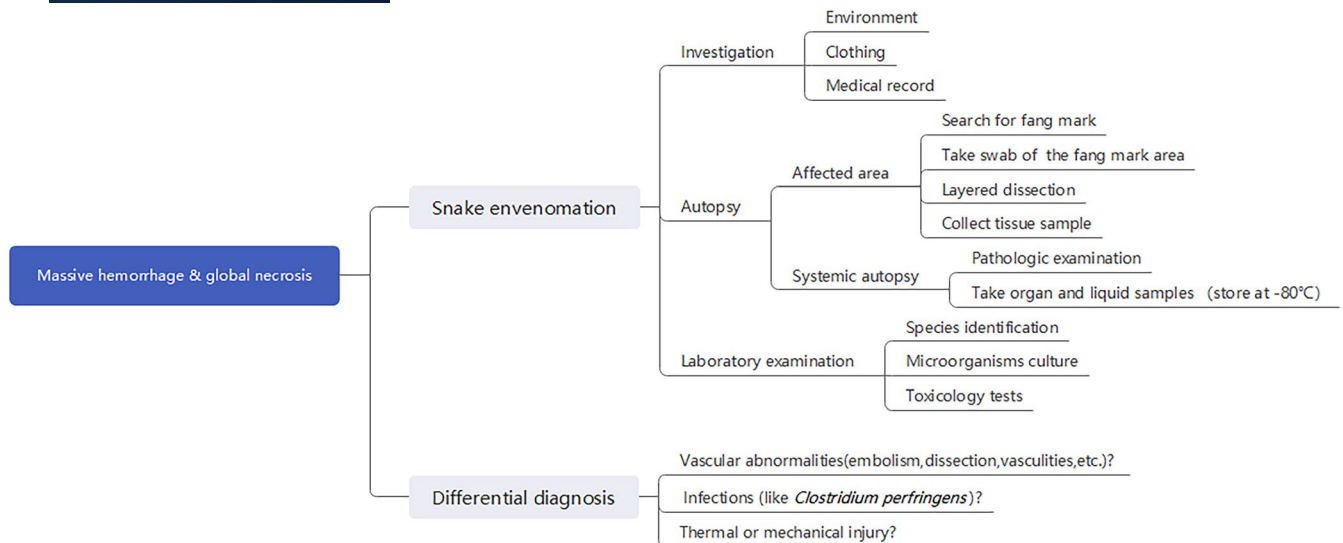


FIGURE 4 Identification of *D. acutus* envenomation case

stejnegeri (4/158) [9]. Therefore, we chose these 4 specific snake venoms when conducting our ELISA testing (Table 2) [5,30].

The ELISA results indicated that the highest concentration of venom was found in the kidneys. Kidneys seem to be an important target organ of *D. acutus* venom. Wei C.B. et al. reported on the pharmacokinetics of AaH I (an active hemorrhagic component in *D. acutus* venom) in New Zealand rabbits, using ^{125}I , and found that the strongest radioactivity was observed in the liver and kidneys [31]. Victims can demonstrate abnormalities in liver and renal function [17]. The concentration of the toxin in the muscles of the left leg was also very high, which supports this area as being the original bite site. In the method we used here, hemolysis would cause a high background OD value resulting in false positive; therefore, blood was not used for the ELISA test in this case due to severe hemolysis. Similarly in other organs, if the congestion was too severe, the result could be negatively affected. In this situation, we were able to identify organ samples that were not severely congested or selected less congested portions of the organs as samples and washed the specimens. In our opinion, the most relevant tissue sample would have been the thigh muscle since the toxicologic effects of *D. acutus* venom were most notable in this portion of muscle, and this was also the initial site of envenomation. No urine was available in this case. No vitreous electrolyte analysis was conducted, but this will be added into our future examinations.

Previously, ELISA results quantifying the concentration of venom in different human organs in *D. acutus* envenomation have rarely been reported. Therefore, our results could serve as a reference for the lethal concentration of *D. acutus* venom in humans.

These pathological changes characterized by massive hemorrhage and global necrosis must be distinguished from vascular abnormalities (embolism, dissection, vasculitis, etc.), microbial infections (such as *Clostridium perfringens*), thermal, or mechanical

injuries (Figure 4). The significant blistering and hemorrhagic bullae are relatively specific symptoms of *D. acutus* envenomation among the mechanisms we considered. Blisters or hemorrhagic bullae have previously been found in 13.26% of *D. acutus* envenomation victims (Table 1). In this case, there were no corresponding abnormalities found during the examination of the blood vessels and heart, and no strongly pathogenic bacteria were found in the microbial cultures (cultures only showed bacterial species resulting from postmortem growth), no history of thermal injury, or mechanical injury. Thus, the corresponding conditions mentioned above could be excluded.

In conclusion, it is important to consider the victim's environment as well as their clothing, which may offer clues to a snakebite history and even help identify the snake species. A thorough medical review of these victims and their clinical characteristics would also help. Consulting with relevant experts is preferable. Extra attention and effort should be given to the identification of fang marks. Layered dissection and extraction of liquid (blood or exudates) and tissues in the suspected area of envenomation is necessary (Figure 4). Our hope is that the autopsy results, histopathological findings, and ELISA results reported here can offer assistance in forensic practice and further research.

ACKNOWLEDGMENTS

We thank Bixun Niu and Siyue Lu for editing the manuscript. We thank Dr. Zhang Yun for offering to assist by acting as a consultant for advice and the ELISA tests.

ETHICAL APPROVAL

This study was approved by the Ethics Committee of Tongji Medical College, Huazhong University of Science and Technology, Wuhan, Hubei, China.

Informed consent: Written informed consent was obtained from the family members of the patient.

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How to cite this article: Huang F, Zhao S, Tong F, et al. Unexpected death in a young man associated with a unilateral swollen leg: Pathological and toxicological findings in a fatal snakebite from *Deinagkistrodon acutus* (Chinese moccasin). *J Forensic Sci*. 2020;00:1–7. <https://doi.org/10.1111/1556-4029.14622>